

YOUNGJIN HONG

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RESEARCH INTERESTS

Vision-Language-Action Models, Multimodal Robot Learning, Generative Policies, Contact-Rich Manipulation

EDUCATION

University of Minnesota (UMN) · Minneapolis, MN, USA

Ph.D. Electrical Engineering · September 2024–Present

Sungkyunkwan University (SKKU) · Suwon, Korea

M.S. / B.S. in Mechanical Engineering · March 2016–February 2024

RESEARCH & INDUSTRY EXPERIENCE

Research Assistant · September 2024–Present

[Choice Robotics Lab](#), University of Minnesota · Minneapolis, MN

- Conduct research on Vision-Language-Action models and computer vision for real-world applications.
- Develop learning-based robotic systems using simulation and physical platforms including MuJoCo and ALOHA2.
- Train multimodal models and conduct large-scale experiments on GPU clusters using PyTorch and SLURM.

Robot Research Engineer · January–July 2024

[Hanwha Aerospace](#), Manned-Unmanned Teaming (MUM-T) Research Center · Seoul, Korea

- Built Isaac Sim environments and supported field testing for autonomous unmanned ground vehicle research.

SELECTED PROJECTS

VLM-Guided Person Re-Identification and Tracking · September 2025–Present

Collaboration with ETRI

- Developed a language-guided person tracking pipeline integrating **YOLOv11-L**, **ByteTrack**, and **LLaVA-NeXT** for identifying and re-identifying a described target in continuous video.
- Designed an **adaptive VLM triggering mechanism** that invokes multimodal reasoning only during initialization, target loss, low-confidence tracking, or periodic verification.

Language–Action Cycles for Learning Manipulation Tasks · June–October 2025

- Proposed a **bidirectional language–action cycle** that jointly learns action-to-language and language-to-action mappings for scalable behavior acquisition.
- Designed a semantic consistency verifier for filtering self-generated samples, enabling reliable self-improvement.

SELECTED PUBLICATIONS / PRESENTATIONS

Youngjin Hong, Tian Xie, Mingen Li, et al. “NUDGE: Efficient Structured Action Correction for Vision-Language-Action Models under Latency.” *Submitted to ICRA 2027*.

Hantao Ye, Ross Worobel, Zhuoli Xie, Mingen Li, Houjian Yu, **Youngjin Hong**, Changhyun Choi. “POINTCAST: One World Model for Rigid, Articulated, and Deformable Object Manipulation.” *Submitted to ICRA 2027*. [\[Website\]](#)

Youngjin Hong*, Houjian Yu* (*equal contribution) et al. “LACY: A Vision-Language Model-based Language-Action Cycle for Self-Improving Robotic Manipulation.” *ICRA 2026*. [\[Website\]](#) [\[Paper\]](#)

Mingen Li, Houjian Yu, Yixuan Huang, **Youngjin Hong** et al. “Hierarchical DLO Routing with Reinforcement Learning and In-Context Vision-language Models.” *ICRA 2026*. **Best Paper Award on Robot Learning Finalist**. [\[Website\]](#) [\[Paper\]](#)

SKILLS

Programming: Python, C++, MATLAB

ML: PyTorch, Hugging Face, Vision-Language Models (VLM), Vision-Language-Action Models (VLA), Diffusion/Flow Matching

Robotics: ROS, MuJoCo, Isaac Gym/Sim, ALOHA2, Franka, MELFA

Infra: Linux, Git, Docker, SLURM, GPU clusters